Lesson 15: Metric Spaces

AI/ML Mathematics

Contents

1 Motivation **2**

2 Metric Spaces **2**

2.1 Definition and First Properties 2

2.2 Examples 2

2.3 Basic Topology 3

3 Convergence and Completeness **4**

3.1 Convergent Sequences 4

3.2 Cauchy Sequences and Completeness 4

3.3 Dense Subsets and Separability 6

4 The Completion Theorem **6**

5 Exercises **8**

1 Motivation

Reinforcement learning algorithms seek fixed points of operators acting on spaces of functions. The Bellman optimality operator

$$(TV)(s) = \max_{a \in \mathcal{A}} \Big[r(s,a) + \gamma \sum_{s'} p(s' \mid s,a)\, V(s') \Big] \tag{1}$$

maps a value function $V \colon \mathcal{S} \to \mathbb{R}$ to another value function TV. To prove that value iteration $V_{n+1} = TV_n$ converges, we need precise answers to four questions:

1. What does "distance between two value functions" mean?
2. What does "V_n converges to V^*" mean?
3. Does the limit V^* actually belong to the same space as the V_n?
4. Is the limit unique?

The theory of *metric spaces* provides the framework for all four. A metric formalizes distance (question 1), convergence of sequences is defined via the metric (question 2), completeness guarantees that Cauchy sequences have limits in the space (question 3), and the Hausdorff property of metric spaces ensures uniqueness (question 4). The Banach contraction mapping theorem – which we prove in a later lesson – ties everything together, but its hypotheses are stated entirely in the language of this lesson.

The Bellman Operator on a Metric Space

The Bellman operator T acts on the space of bounded real-valued functions $b(\mathcal{S})$ equipped with the supremum metric $d_\infty(V,W) = \sup_{s \in \mathcal{S}} |V(s) - W(s)|$. One shows that $d_\infty(TV, TW) \le \gamma\, d_\infty(V,W)$ for any discount factor $\gamma \in [0,1)$, making T a *contraction*. The contraction mapping theorem then yields existence, uniqueness, and geometric convergence of value iteration – but only if $(b(\mathcal{S}), d_\infty)$ is a *complete* metric space. Every concept in this lesson (metric, convergence, Cauchy sequence, completeness) is indispensable to that argument.

2 Metric Spaces

2.1 Definition and First Properties

Definition 2.1 (Metric Space)**.** A *metric* on a set X is a function $d \colon X \times X \to [0,\infty)$ satisfying, for all $x, y, z \in X$:

(M1) $d(x,y) = 0$ if and only if $x = y$. (identity of indiscernibles)
(M2) $d(x,y) = d(y,x)$. (symmetry)
(M3) $d(x,z) \le d(x,y) + d(y,z)$. (triangle inequality)

The pair (X,d) is called a *metric space*.

Remark 2.2. Non-negativity $d(x,y) \ge 0$ follows from the axioms and need not be postulated separately. Indeed, by (M3) and (M2), $0 = d(x,x) \le d(x,y) + d(y,x) = 2\, d(x,y)$, so $d(x,y) \ge 0$.

2.2 Examples

Example 2.3 (Euclidean Metric on $\mathbb{R}^n$)**.** On $X = \mathbb{R}^n$, define $d_2(x,y) = \big(\sum_{i=1}^n (x_i - y_i)^2\big)^{1/2}$. Axioms (M1) and (M2) are immediate. The triangle inequality follows from the Cauchy–Schwarz inequality: for $u, v \in \mathbb{R}^n$, $\big(\sum_i u_i v_i\big)^2 \le \big(\sum_i u_i^2\big)\big(\sum_i v_i^2\big)$, which implies $\|u+v\|_2 \le \|u\|_2 + \|v\|_2$ by expanding $\|u+v\|_2^2 = \|u\|_2^2 + 2\langle u, v\rangle + \|v\|_2^2 \le (\|u\|_2 + \|v\|_2)^2$.

Example 2.4 (p-Metrics on $\mathbb{R}^n$)**.** For $1 \le p \le \infty$, define $d_p(x,y) = \|x-y\|_p$ where

$$\|x\|_p = \begin{cases} \left(\sum_{i=1}^n |x_i|^p\right)^{1/p} & \text{if } 1 \le p < \infty, \\ \max_{1\le i\le n} |x_i| & \text{if } p = \infty. \end{cases}$$

Each d_p is a metric. For $1 < p < \infty$, the triangle inequality is Minkowski's inequality (proved via Hölder's inequality). The case $p = \infty$ is straightforward: $\max_i |x_i + y_i| \le \max_i(|x_i| + |y_i|) \le \max_i |x_i| + \max_i |y_i|$.

Example 2.5 (Discrete Metric)**.** On any set X, define $d(x,y) = 0$ if $x = y$ and $d(x,y) = 1$ if $x \ne y$. All three axioms are easily verified; in particular, the triangle inequality holds because $d(x,z) \le 1 \le d(x,y) + d(y,z)$ whenever $x \ne z$ (at least one of $d(x,y), d(y,z)$ equals 1). Every subset of X is open: for any $x \in A \subseteq X$, $B(x, 1/2) = \{x\} \subseteq A$.

Example 2.6 (The Sup Metric on $C[a,b]$)**.** Let $C[a,b]$ denote the set of continuous real-valued functions on $[a,b]$. Define

$$d_\infty(f,g) = \max_{t\in[a,b]} |f(t) - g(t)|.$$

The maximum exists by the extreme value theorem (continuous functions on compact sets attain their extrema). Axioms (M1)–(M3) follow from the corresponding properties of $|\cdot|$ on $\mathbb{R}$.

One can also define $d_1(f,g) = \int_a^b |f(t) - g(t)|\,dt$, which is a different metric on $C[a,b]$. These two metrics induce different topologies and have different completeness properties (see Section 3).

Example 2.7 (Sequence Spaces ℓ^p)**.** For $1 \le p < \infty$, define

$$\ell^p = \left\{x = (x_n)_{n=1}^\infty \subset \mathbb{R} : \sum_{n=1}^\infty |x_n|^p < \infty\right\}, \quad d_p(x,y) = \left(\sum_{n=1}^\infty |x_n - y_n|^p\right)^{1/p}.$$

For $p = \infty$, $\ell^\infty = \{x = (x_n) : \sup_n |x_n| < \infty\}$ with $d_\infty(x,y) = \sup_n |x_n - y_n|$. The triangle inequality for ℓ^p ($1 < p < \infty$) is the infinite-sum version of Minkowski's inequality.

2.3 Basic Topology

Definition 2.8 (Open Ball)**.** In a metric space (X,d), the *open ball* of radius $r > 0$ centered at $x \in X$ is

$$B(x,r) = \{y \in X : d(x,y) < r\}.$$

The *closed ball* is $\overline{B}(x,r) = \{y \in X : d(x,y) \le r\}$.

Definition 2.9 (Open and Closed Sets)**.** Let (X,d) be a metric space.

(i) A set $G \subseteq X$ is *open* if for every $x \in G$ there exists $r > 0$ such that $B(x,r) \subseteq G$.

(ii) A set $F \subseteq X$ is *closed* if $X \setminus F$ is open.

Proposition 2.10 (Properties of Open and Closed Sets)**.** *In a metric space (X,d):*

(a) $\emptyset$ and X are both open and closed.

(b) Any union of open sets is open.

(c) Any finite intersection of open sets is open.

(d) Any intersection of closed sets is closed.

(e) Any finite union of closed sets is closed.

(f) Every open ball $B(x,r)$ is an open set.

Proof. **(a)** $\emptyset$ is open vacuously (no point to check). X is open because $B(x,r) \subseteq X$ for every $x \in X$ and $r > 0$. Their complements are X and $\emptyset$, so both are also closed.

(b) Let $\{G_\alpha\}_{\alpha\in I}$ be open sets and $x \in \bigcup_\alpha G_\alpha$. Then $x \in G_\beta$ for some $\beta \in I$, so there exists $r > 0$ with $B(x,r) \subseteq G_\beta \subseteq \bigcup_\alpha G_\alpha$.

(c) Let $G_1, \ldots, G_n$ be open and $x \in \bigcap_{i=1}^n G_i$. For each i, choose $r_i > 0$ with $B(x, r_i) \subseteq G_i$. Set $r = \min(r_1, \ldots, r_n) > 0$ (finite minimum of positive numbers). Then $B(x, r) \subseteq B(x, r_i) \subseteq G_i$ for all i, so $B(x, r) \subseteq \bigcap_{i=1}^n G_i$.

(d)–(e) These follow from (b)–(c) by De Morgan's laws: $X \setminus \bigcap_\alpha F_\alpha = \bigcup_\alpha (X \setminus F_\alpha)$ and $X \setminus \bigcup_{i=1}^n F_i = \bigcap_{i=1}^n (X \setminus F_i)$.

(f) Let $y \in B(x, r)$ and set $\varepsilon = r - d(x, y) > 0$. For any $z \in B(y, \varepsilon)$, the triangle inequality gives $d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + \varepsilon = r$, so $z \in B(x, r)$. Hence $B(y, \varepsilon) \subseteq B(x, r)$. □

Definition 2.11 (Interior, Closure, Boundary)**.** Let $A \subseteq X$ in a metric space (X, d).

(i) The *interior* of A is $\operatorname{int}(A) = \bigcup\{G \subseteq A : G \text{ is open}\}$, the largest open set contained in A.
(ii) The *closure* of A is $\operatorname{cl}(A) = \bigcap\{F \supseteq A : F \text{ is closed}\}$, the smallest closed set containing A.
(iii) The *boundary* of A is $\partial A = \operatorname{cl}(A) \setminus \operatorname{int}(A)$.

Proposition 2.12 (Closure Characterization)**.** *Let (X, d) be a metric space and $A \subseteq X$. The following are equivalent:*

(i) $x \in \operatorname{cl}(A)$.
(ii) For every $r > 0$, $B(x, r) \cap A \neq \emptyset$.
(iii) There exists a sequence (a_n) in A with $a_n \to x$.

Proof. **(i) ⇒ (ii).** Suppose $x \in \operatorname{cl}(A)$ but there exists $r > 0$ with $B(x, r) \cap A = \emptyset$. Then $A \subseteq X \setminus B(x, r)$. Since $B(x, r)$ is open, $X \setminus B(x, r)$ is closed. By definition of closure, $\operatorname{cl}(A) \subseteq X \setminus B(x, r)$. But $x \in B(x, r)$, so $x \notin X \setminus B(x, r)$, contradicting $x \in \operatorname{cl}(A)$.

(ii) ⇒ (iii). For each $n \in \mathbb{N}$, choose $a_n \in B(x, 1/n) \cap A$. Then $d(a_n, x) < 1/n \to 0$, so $a_n \to x$.

(iii) ⇒ (i). Let (a_n) be a sequence in A with $a_n \to x$. Let F be any closed set with $A \subseteq F$. Then each $a_n \in F$. Since F is closed, $X \setminus F$ is open. If $x \notin F$, then $x \in X \setminus F$, so there exists $r > 0$ with $B(x, r) \subseteq X \setminus F$. But $a_n \to x$ means $a_n \in B(x, r)$ for large n, giving $a_n \notin F$, contradicting $a_n \in A \subseteq F$. Hence $x \in F$. Since this holds for every closed $F \supseteq A$, we get $x \in \operatorname{cl}(A)$. □

3 Convergence and Completeness

3.1 Convergent Sequences

Definition 3.1 (Convergence)**.** A sequence (x_n) in a metric space (X, d) *converges* to $x \in X$ if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon$ for all $n \geq N$. We write $x_n \to x$ or $\lim_{n\to\infty} x_n = x$.

Proposition 3.2 (Uniqueness of Limits)**.** *In a metric space, limits are unique: if $x_n \to x$ and $x_n \to y$, then $x = y$.*

Proof. By the triangle inequality, $0 \leq d(x, y) \leq d(x, x_n) + d(x_n, y)$ for every n. Given $\varepsilon > 0$, choose N_1 with $d(x_n, x) < \varepsilon/2$ for $n \geq N_1$ and N_2 with $d(x_n, y) < \varepsilon/2$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$, $d(x, y) < \varepsilon/2 + \varepsilon/2 = \varepsilon$. Since $\varepsilon > 0$ was arbitrary and $d(x, y) \geq 0$, we conclude $d(x, y) = 0$, hence $x = y$ by (M1). □

3.2 Cauchy Sequences and Completeness

Definition 3.3 (Cauchy Sequence)**.** A sequence (x_n) in (X, d) is a *Cauchy sequence* if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $d(x_m, x_n) < \varepsilon$ for all $m, n \geq N$.

Proposition 3.4 (Convergent Implies Cauchy)**.** *Every convergent sequence in a metric space is Cauchy.*

Proof. Suppose $x_n \to x$. Given $\varepsilon > 0$, choose N with $d(x_n, x) < \varepsilon/2$ for all $n \geq N$. For $m, n \geq N$,

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \qquad \square$$

Cauchy Does Not Imply Convergent

The converse fails in incomplete spaces. Consider $\mathbb{Q}$ with the usual metric $d(x, y) = |x-y|$. Define $x_1 = 1$ and $x_{n+1} = (x_n + 2/x_n)/2$. Each $x_n \in \mathbb{Q}$, and one shows by induction that (x_n) is Cauchy. The sequence converges to $\sqrt{2}$ in $\mathbb{R}$, but $\sqrt{2} \notin \mathbb{Q}$. So (x_n) is Cauchy in $\mathbb{Q}$ with no limit in $\mathbb{Q}$.

Definition 3.5 (Complete Metric Space)**.** A metric space (X, d) is *complete* if every Cauchy sequence in X converges to a limit in X.

Theorem 3.6 (Completeness of $\mathbb{R}$)**.** $(\mathbb{R}, |\cdot|)$ *is a complete metric space.*

Proof. Let (x_n) be a Cauchy sequence in $\mathbb{R}$.

Step 1: (x_n) is bounded. Choose N with $|x_m - x_n| < 1$ for all $m, n \geq N$. Then for $n \geq N$, $|x_n| \leq |x_N| + 1$. Setting $M = \max(|x_1|, \ldots, |x_{N-1}|, |x_N| + 1)$, we have $|x_n| \leq M$ for all n.

Step 2: Extract a convergent subsequence. By the Bolzano–Weierstrass theorem, every bounded sequence in $\mathbb{R}$ has a convergent subsequence. Thus there exist a subsequence (x_{n_k}) and a point $x \in \mathbb{R}$ with $x_{n_k} \to x$.

Step 3: The full sequence converges. Given $\varepsilon > 0$, choose N_1 with $|x_m - x_n| < \varepsilon/2$ for $m, n \geq N_1$, and K with $|x_{n_k} - x| < \varepsilon/2$ for $k \geq K$. For $n \geq N_1$, pick $k \geq K$ with $n_k \geq N_1$. Then

$$|x_n - x| \leq |x_n - x_{n_k}| + |x_{n_k} - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $x_n \to x$. $\square$

Theorem 3.7 (Completeness of $\mathbb{R}^n$)**.** $(\mathbb{R}^n, d_p)$ *is a complete metric space for every* $1 \leq p \leq \infty$.

Proof. Let $(x^{(k)})_{k=1}^{\infty}$ be Cauchy in $(\mathbb{R}^n, d_p)$, with $x^{(k)} = (x_1^{(k)}, \ldots, x_n^{(k)})$.

Coordinate-wise convergence. For each $i \in \{1, \ldots, n\}$,

$$|x_i^{(k)} - x_i^{(m)}| \leq \|x^{(k)} - x^{(m)}\|_p,$$

since the ℓ^p norm dominates each coordinate. Hence $(x_i^{(k)})_{k=1}^{\infty}$ is Cauchy in $\mathbb{R}$. By Theorem 3.6, $x_i^{(k)} \to x_i$ for some $x_i \in \mathbb{R}$.

Convergence in d_p. Let $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$. Given $\varepsilon > 0$, choose K with $\|x^{(k)} - x^{(m)}\|_p < \varepsilon/2$ for $k, m \geq K$. For any finite set of coordinates and $k \geq K$, pass $m \to \infty$ in $|x_i^{(k)} - x_i^{(m)}|$ to obtain $|x_i^{(k)} - x_i| \leq \varepsilon/2$. Thus $\|x^{(k)} - x\|_p \leq \varepsilon/2 < \varepsilon$ for $k \geq K$, and $x^{(k)} \to x$ in $(\mathbb{R}^n, d_p)$. $\square$

Completeness of $\mathbb{R}^n$

The spaces $(\mathbb{R}^n, d_p)$ for $1 \leq p \leq \infty$ are all complete. In RL with a finite state space $|\mathcal{S}| = n$, value functions are vectors in $\mathbb{R}^n$ and the space of all value functions is complete under any p-metric. This is one reason finite-state MDPs are analytically tractable.

3.3 Dense Subsets and Separability

Definition 3.8 (Dense Subset, Separable Space)**.** Let (X, d) be a metric space.

(i) A subset $D \subseteq X$ is *dense* in X if $\mathrm{cl}(D) = X$, equivalently, if for every $x \in X$ and every $r > 0$, $B(x, r) \cap D \neq \emptyset$.

(ii) (X, d) is *separable* if it contains a countable dense subset.

Example 3.9. (a) $\mathbb{R}^n$ is separable: $\mathbb{Q}^n$ is countable and dense (every open ball in $\mathbb{R}^n$ contains a point with all rational coordinates).

(b) $(C[a, b], d_\infty)$ is separable. By the Weierstrass approximation theorem, every continuous function on $[a, b]$ can be uniformly approximated by polynomials. Polynomials with rational coefficients form a countable dense subset.

(c) ℓ^p is separable for $1 \leq p < \infty$: the set of sequences with finitely many nonzero rational entries is countable and dense.

(d) ℓ^∞ is *not* separable. The uncountable collection $\{x_A : A \subseteq \mathbb{N}\}$ where $(x_A)_n = \mathbf{1}_A(n)$ satisfies $d_\infty(x_A, x_B) = 1$ for $A \neq B$. Any dense subset must contain a point in each ball $B(x_A, 1/3)$, and these balls are pairwise disjoint, so the dense subset is uncountable.

4 The Completion Theorem

The example of $\mathbb{Q}$ inside $\mathbb{R}$ suggests a general construction: given an incomplete metric space, we can "fill in the gaps" by adding limits of Cauchy sequences. The following theorem makes this precise.

Theorem 4.1 (Completion of a Metric Space)**.** *Let (X, d) be a metric space. There exists a complete metric space $(\hat{X}, \hat{d})$ and an isometry $\iota\colon X \to \hat{X}$ (meaning $\hat{d}(\iota(x), \iota(y)) = d(x, y)$ for all $x, y \in X$) such that $\iota(X)$ is dense in $\hat{X}$. Moreover, the completion is unique up to isometry: if $(\hat{X}_1, \hat{d}_1, \iota_1)$ and $(\hat{X}_2, \hat{d}_2, \iota_2)$ are two completions of (X, d), then there exists a surjective isometry $\Phi\colon \hat{X}_1 \to \hat{X}_2$ with $\Phi \circ \iota_1 = \iota_2$.*

Proof. We break the proof into five steps.

Step 1: Construction of $\hat{X}$. Let $\mathcal{C}$ denote the set of all Cauchy sequences in (X, d). Define an equivalence relation on $\mathcal{C}$ by

$$(x_n) \sim (y_n) \quad \iff \quad \lim_{n\to\infty} d(x_n, y_n) = 0.$$

We verify that $\sim$ is an equivalence relation. Reflexivity: $d(x_n, x_n) = 0 \to 0$. Symmetry: $d(x_n, y_n) = d(y_n, x_n)$ by (M2). Transitivity: if $(x_n) \sim (y_n)$ and $(y_n) \sim (z_n)$, then $d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n) \to 0$ by (M3).

Let $\hat{X} = \mathcal{C}/\!\sim$ and write $[(x_n)]$ for the equivalence class of (x_n).

Step 2: The metric $\hat{d}$ is well defined. For Cauchy sequences (x_n) and (y_n), the real sequence $(d(x_n, y_n))$ is Cauchy in $\mathbb{R}$: by the triangle inequality,

$$|d(x_m, y_m) - d(x_n, y_n)| \leq d(x_m, x_n) + d(y_m, y_n) \to 0 \quad \text{as } m, n \to \infty.$$

Since $\mathbb{R}$ is complete (Theorem 3.6), the limit $\lim_{n\to\infty} d(x_n, y_n)$ exists. Define

$$\hat{d}\big([(x_n)], [(y_n)]\big) = \lim_{n\to\infty} d(x_n, y_n).$$

Independence of representatives. If $(x_n) \sim (x'_n)$ and $(y_n) \sim (y'_n)$, then

$$|d(x_n, y_n) - d(x'_n, y'_n)| \leq d(x_n, x'_n) + d(y_n, y'_n) \to 0,$$

so the limits agree.

$\hat{d}$ is a metric. (M2) and non-negativity are inherited from d. For (M1): $\hat{d}([(x_n)],[(y_n)]) = 0$ iff $d(x_n, y_n) \to 0$ iff $(x_n) \sim (y_n)$ iff $[(x_n)] = [(y_n)]$. For (M3): $\hat{d}([(x_n)],[(z_n)]) = \lim d(x_n, z_n) \leq \lim[d(x_n, y_n) + d(y_n, z_n)] = \hat{d}([(x_n)],[(y_n)]) + \hat{d}([(y_n)],[(z_n)])$.

Step 3: ι is an isometric embedding and $\iota(X)$ is dense. Define $\iota\colon X \to \hat{X}$ by $\iota(x) = [(x, x, x, \ldots)]$ (the class of the constant sequence at x). Then $\hat{d}(\iota(x), \iota(y)) = \lim_{n\to\infty} d(x, y) = d(x, y)$, so ι is an isometry.

For density: let $p = [(x_n)] \in \hat{X}$ and $\varepsilon > 0$. Since (x_n) is Cauchy, there exists N with $d(x_m, x_n) < \varepsilon/2$ for $m, n \geq N$. Then

$$\hat{d}(p,\, \iota(x_N)) = \lim_{n\to\infty} d(x_n, x_N) \leq \frac{\varepsilon}{2} < \varepsilon.$$

Hence $\iota(X)$ is dense in $\hat{X}$.

Step 4: $\hat{X}$ is complete. Let $(p_k)_{k=1}^{\infty}$ be a Cauchy sequence in $(\hat{X}, \hat{d})$. Since $\iota(X)$ is dense, for each k choose $z_k \in X$ with $\hat{d}(p_k, \iota(z_k)) < 1/k$.

(z_k) is Cauchy in X. For $k, m \in \mathbb{N}$,

$$\begin{aligned} d(z_k, z_m) &= \hat{d}(\iota(z_k), \iota(z_m)) \\ &\leq \hat{d}(\iota(z_k), p_k) + \hat{d}(p_k, p_m) + \hat{d}(p_m, \iota(z_m)) \\ &< \frac{1}{k} + \hat{d}(p_k, p_m) + \frac{1}{m}. \end{aligned}$$

Since (p_k) is Cauchy in $\hat{X}$, the right-hand side tends to 0 as $k, m \to \infty$. Hence (z_k) is Cauchy in X.

(p_k) converges to $p = [(z_k)] \in \hat{X}$. We have

$$\hat{d}(p_k, p) \leq \hat{d}(p_k, \iota(z_k)) + \hat{d}(\iota(z_k), p).$$

The first term is at most $1/k \to 0$. For the second term,

$$\hat{d}(\iota(z_k), p) = \hat{d}(\iota(z_k),\, [(z_j)]) = \lim_{j\to\infty} d(z_k, z_j).$$

Since (z_j) is Cauchy, $\lim_{j\to\infty} d(z_k, z_j) \to 0$ as $k \to \infty$ (given $\varepsilon > 0$, choose K with $d(z_j, z_m) < \varepsilon$ for $j, m \geq K$; then for $k \geq K$, $\lim_j d(z_k, z_j) \leq \varepsilon$). Therefore $\hat{d}(p_k, p) \to 0$, and $(\hat{X}, \hat{d})$ is complete.

Step 5: Uniqueness up to isometry. Let $(\hat{X}_1, \hat{d}_1, \iota_1)$ and $(\hat{X}_2, \hat{d}_2, \iota_2)$ be two completions of (X, d). Define $\varphi\colon \iota_1(X) \to \hat{X}_2$ by $\varphi(\iota_1(x)) = \iota_2(x)$. Then

$$\hat{d}_2(\varphi(\iota_1(x)),\, \varphi(\iota_1(y))) = \hat{d}_2(\iota_2(x), \iota_2(y)) = d(x, y) = \hat{d}_1(\iota_1(x), \iota_1(y)),$$

so φ is an isometry from $\iota_1(X)$ to $\iota_2(X)$. In particular, φ is uniformly continuous. Since $\iota_1(X)$ is dense in $\hat{X}_1$ and $\hat{X}_2$ is complete, φ extends uniquely to a uniformly continuous map $\Phi\colon \hat{X}_1 \to \hat{X}_2$ (by the extension theorem for uniformly continuous maps into complete spaces). By continuity, Φ preserves the isometry property: $\hat{d}_2(\Phi(p), \Phi(q)) = \hat{d}_1(p, q)$ for all $p, q \in \hat{X}_1$.

Finally, Φ is surjective: $\Phi(\hat{X}_1) \supseteq \Phi(\iota_1(X)) = \iota_2(X)$, and $\Phi(\hat{X}_1)$ is complete (as an isometric image of a complete space), hence closed. Since $\iota_2(X)$ is dense in $\hat{X}_2$, we get $\hat{X}_2 = \mathrm{cl}(\iota_2(X)) \subseteq \mathrm{cl}(\Phi(\hat{X}_1)) = \Phi(\hat{X}_1)$. Thus Φ is a surjective isometry with $\Phi \circ \iota_1 = \iota_2$. □

Completion and Function Approximation

In RL with function approximation, we often work with a parametric class of value functions (e.g., linear combinations of features, or neural network outputs) that may not be complete under the sup norm. The completion theorem guarantees that this class can be isometrically embedded into a complete space. Understanding the structure of the com-

pleted space helps determine whether approximate value iteration converges and what it converges to.

5 Exercises

Exercise 5.1 (Metrics from Metrics)**.** Let (X, d) be a metric space. Prove that the following are also metrics on X:

(a) $d_1(x, y) = \min(d(x, y), 1)$.

(b) $d_2(x, y) = \dfrac{d(x, y)}{1 + d(x, y)}$.

For each, determine whether (X, d_i) is complete whenever (X, d) is complete.

Hint: For (b), show that $f(t) = t/(1 + t)$ is increasing and subadditive on $[0, \infty)$.

Exercise 5.2 (Closed Balls Need Not Be Closures of Open Balls)**.** In a general metric space, the closed ball $\overline{B}(x, r) = \{y : d(x, y) \leq r\}$ may differ from the closure $\mathrm{cl}(B(x, r))$ of the open ball. Give an example by considering the discrete metric on a set with at least two points, and verify explicitly that $\mathrm{cl}(B(x, 1)) \neq \overline{B}(x, 1)$ for a suitable choice of r.

Exercise 5.3 (Completeness of ℓ^∞)**.** Prove that (ℓ^∞, d_∞) is a complete metric space. That is, show that every Cauchy sequence of bounded sequences converges to a bounded sequence.

Hint: Follow the structure of the proof of Theorem 3.7: use coordinate-wise convergence, then verify that the limit is bounded and that convergence holds in d_∞.

Exercise 5.4 (Density of $\mathbb{Q}$ in $\mathbb{R}$)**.** Prove that $\mathbb{Q}$ is dense in $(\mathbb{R}, |\cdot|)$ using only the Archimedean property of $\mathbb{R}$ (for any $x > 0$, there exists $n \in \mathbb{N}$ with $n > x$). That is, show that for every $x \in \mathbb{R}$ and every $\varepsilon > 0$, there exists $q \in \mathbb{Q}$ with $|x - q| < \varepsilon$.

Exercise 5.5 (The Completion of $\mathbb{Q}$)**.** Let $(\hat{\mathbb{Q}}, \hat{d})$ be the completion of $(\mathbb{Q}, |\cdot|)$ as constructed in Theorem 4.1. Show that $\hat{\mathbb{Q}}$ is isometric to $(\mathbb{R}, |\cdot|)$.

Hint: Use uniqueness (Step 5 of the proof). It suffices to show that $\mathbb{R}$ is complete and that $\mathbb{Q}$ embeds isometrically as a dense subset of $\mathbb{R}$.

References